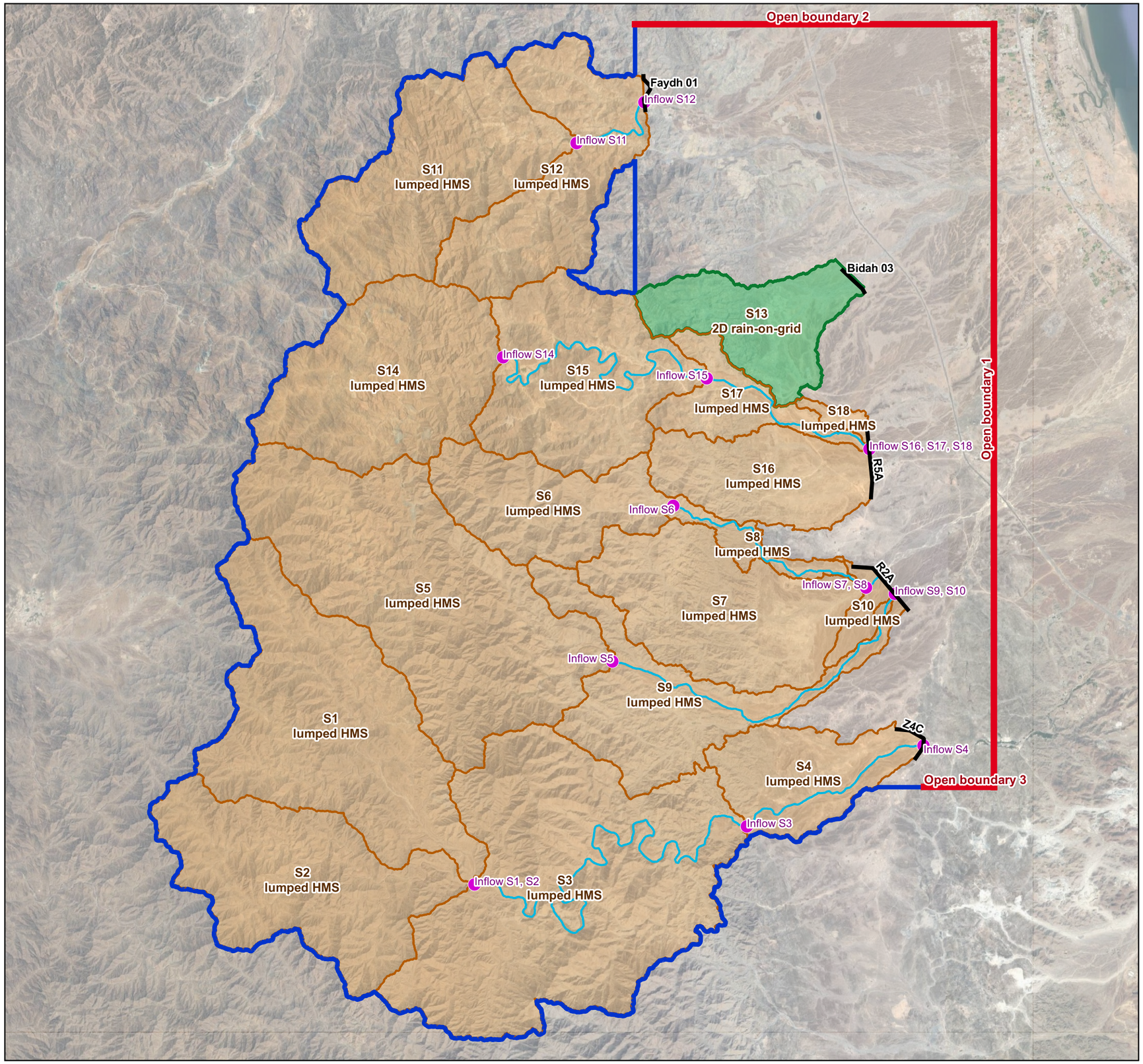
# Wadi2D hybrid model, Block 2: setup

Lumped HMS hills feeding the 2D model downstream




## Legend

Subbasins

- Subbasin, lumped HMS (rain masked in 2D)
- Subbasin, 2D rain-on-grid
- 2D model domain (closed walls except open boundaries)
- Open boundary
- Lumped hydrograph enters 2D here
- Reach, routed by the 2D model here
- Dam section

How the model is built

Hills: 17 subbasins, 578 km2, run in lumped HMS (SCS CN, lag). Their rain is masked out of the 2D model.

Each hydrograph enters the 2D model at its magenta point, 12 points.

2D: S13 and the plain, rain-on-grid on 20 m cells with 5 m terrain tables, CN grid, Manning n 0.04.

Water leaves through 3 open boundaries, normal depth at S 0.008. Every other edge of the domain is a closed wall. HMS reaches are not routed: the 2D model carries the hydrographs down the wadis.

Storm: Block 2 200-yr, 251 mm, 24 h, storm area 300 km2. Simulated 48 h.

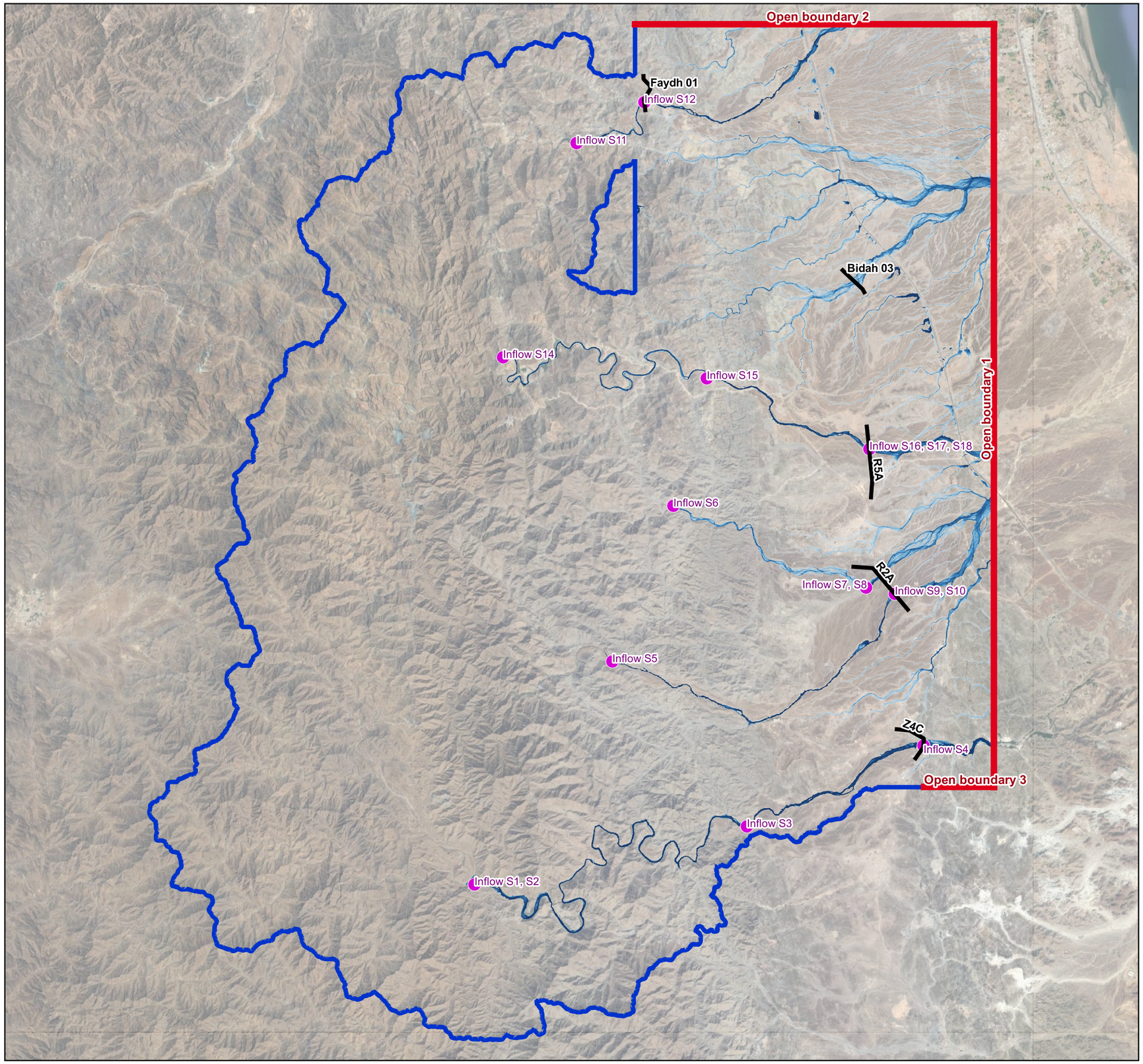
Water: 103.3 Mm3 from the lumped hills, 39.1 Mm3 of rain on the 2D area after losses, 139.0 Mm3 out. Mass error 2e-08.

0 2.5 5 7.5 10 km

EPSG:32640. Imagery: Google.

# Wadi2D hybrid model, Block 2: maximum depth

Block 2 200-yr storm. Depth at the 5 m pixel, sloped water surface




## Legend

Maximum depth, m

3

0

2D model domain (closed walls except open boundaries)

Open boundary

Lumped hydrograph enters 2D here

Dam section

Peak flow at the dam sections

Faydh 01: 849 m3/s at hour 13.3
Bidah 03: 668 m3/s at hour 12.3
R5A: 1,030 m3/s at hour 14.1
R2A: 1,545 m3/s at hour 13.7
Z4C: 2,040 m3/s at hour 14.9

Water: 103.3 Mm3 from the lumped hills, 39.1 Mm3 of rain on the 2D area after losses, 139.0 Mm3 out. Mass error 2e-08.

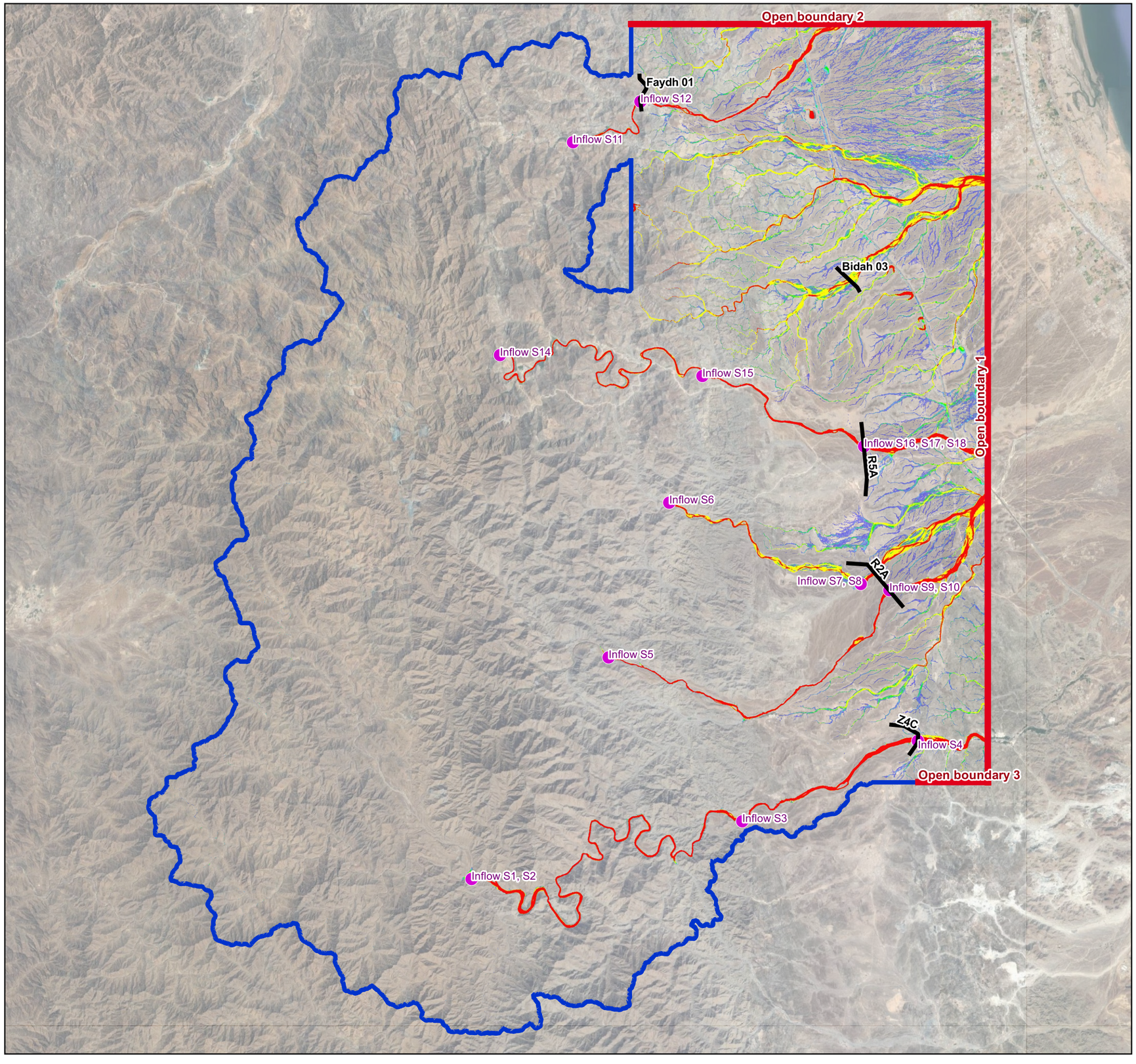
Not a design result. No gauge, no culverts, 20 m cells.

0 2.5 5 7.5 10 km

EPSG:32640. Imagery: Google.

# Wadi2D hybrid model, Block 2: AIDR hazard H1 to H6

Block 2 200-yr storm. Pixel depth x interpolated velocity, 15-minute maps




## Legend

AIDR hazard class

- H1
- H2
- H3
- H4
- H5
- H6
- 2D model domain (closed walls except open boundaries)
- Open boundary
- Lumped hydrograph enters 2D here
- Dam section

Area by class, 47.6 km2 in all

H1: 14.4 km2
H2: 6.4 km2
H3: 1.0 km2
H4: 4.0 km2
H5: 11.6 km2
H6: 10.1 km2

AIDR limits and rule as the Majlas dam-break study: a pixel takes the highest class any limit gives.

Not a design result. No gauge, no culverts, 20 m cells.

0 2.5 5 7.5 10 km

EPSG:32640. Imagery: Google.